Week 1 Meet your robot

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

Robot challenge A robot welcome signal

Predict together Number these actions in run order: drive, show Y, wait. What changes if the first and last lines trade places?

Our prediction _____

```
await signal("Y")
await pause(500)
await drive(90)
```

Read this fragment. Use the full project supplied by your mentor.

Run	One change	Prediction	Observed result
1			
2			
3			

Build A removable robot mascot

Hand-test result _____ One change we made _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What did you learn about instruction order? What did you notice when the robot ran?

Next idea _____ Saved as _____

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Week 1 build card

A removable robot mascot

Team ____ A ____ B ____ C ____

Make a name flag or face that stays upright and attaches without covering a wheel, port, button or cable.

Materials: Small Technic beams and pins, pre-cut card, crayons, removable tape and a mentor-made mounting template.

Try it

1. Choose a team name and lay out a lightweight mast or face.

2. Build a wide or braced support using the mounting template.

3. Gently tilt and tap the model by hand. Improve one weak connection.

Our jobs

Builder connects. Designer chooses parts and checks the plan. Tester hand-tests and describes the result. Everyone builds. Rotate when the mentor calls time.

Test and improve

Stands after a gentle tap; stays inside the clearance template; no loose parts.

We noticed _____. We changed _____.

Fit the approved mascot to the stopped base for the retest. The first robot teams use a mentor example or the bare base.

Fit only to a stopped robot after the mentor checks clearance. Leave wheels, buttons, cables and sensors clear.

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Session attendance and delivery record

Session number _____ Date _____ Actual start _____ Actual end _____

Adult lead _____ Mentors _____

Student ID	Minutes present	Language used	Driver turn confirmed	Notes
S01				
S02				
S03				
S04				
S05				
S06				
S07				
S08				
S09				
S10				
S11				
S12				

Use language labels Python, Word Blocks, or Unplugged. A short observation can identify one concept explained or an access adjustment. Keep medical and family information out of this sheet.

Unique students today _____ Sum of participant minutes _____ / 60 = contact hours _____

Mentor preparation hours _____ Mentor delivery hours _____ Adult hours _____

What worked and what should change next week

_____

_____

_____

Equipment or file issue to resolve before next session

_____

_____

Robot jobs

Print two copies of this page. Cut between the three jobs. Rotate at the mentor’s timer.

Programmer / driver

Edits the agreed letter, value or short recipe. Says the planned change aloud. Starts only after the navigator confirms the lane. Uses the demonstrated Stop control. Does not reach for a moving robot.

Robot handler / navigator

Reads the next step and says a prediction. With the program stopped, positions the robot and payload at the start, fits the approved attachment and resets between runs. Checks the lane and calls “clear.” Keeps hands out during motion.

Test observer

Watches the run and reads the ruler, display or sensor result. Says what actually happened. Records one number, tick or short phrase on the team sheet, or dictates it to the mentor. Suggests one next test.

Build jobs

Print two copies of this page. Cut between the three jobs. Rotate at the mentor’s timer.

Builder

Leads the current assembly step and connects the pieces. All three contribute parts; this is not permission to build everything alone.

Designer / parts manager

Reads the challenge, chooses a simple plan with the team, finds and counts parts, and checks size and mounting limits against the template.

Tester / reporter

Performs the hand test, checks stability and clearance, describes the result, and records a tick or a few words. Suggests one change for the next builder.

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SPIKE Prime hub lights

Check which button, which color and whether the light flashes. A letter on the display is separate program output.

Center button — Green, steady

Hub is on, its OS is compatible with SPIKE, and battery is at least 20%. Still check the correct hub connection, program and clear lane before running.

Center button — White, steady

Hub OS needs an update to work with SPIKE. Ask the mentor to follow the app's update instructions.

Center button — Orange, flashing

Battery is below 20%. Tell the mentor and connect the USB cable to charge.

Center button — Red, flashing

Extended load. Stop, tell the mentor, switch the hub off and let it cool for 15 minutes.

Center button — Purple, flashing

Firmware update is in progress. Let the update finish and follow the app instructions.

Connection button — Blue, flashing

Waiting for a Bluetooth connection. Check the intended hub name with the mentor.

Connection button — Blue, steady

Connected by Bluetooth. This is not a battery-level indication.

LEGO Education reference supplied October 25, 2026. Full source and original image: 06_References/SPIKE_Hub_Color_Guide.md.

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